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Job description parsing with explainable transformer based ensemble models to extract the technical and non-technical skills

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ABSTRACT

The rapid digitization of the economy is transforming the job market, creating new roles and reshaping existing ones. As skill requirements evolve, identifying essential competencies becomes increasingly critical. This paper introduces a novel ensemble model that combines traditional and transformer-based neural networks to extract both technical and non-technical skills from job descriptions. A substantial dataset of job descriptions from reputable platforms was meticulously annotated for 22 IT roles. The model demonstrated superior performance in extracting both non-technical (67% F-score) and technical skills (72% F-score) compared to conventional CRF and hybrid deep learning models. Specifically, the proposed model outperformed these baselines by an average margin of 10% and 6%, respectively, for non-technical skills, and 29% and 6.8% for technical skills. A 5×2cv paired t-test confirmed the statistical significance of these improvements. In addition, to enhance model interpretability, Local Interpretable Model-Agnostic Explanations (LIME) were employed in the experiments.

1. Introduction

The evolution of the internet has profoundly influenced how individuals prepare for their future careers, encompassing the acquisition of skills essential for specific occupations. A skill can be defined as a collection of abilities or behaviors that individuals develop over time through intentional, systematic, and disciplined efforts, along with repetitive practice. These acquired skills empower individuals to effectively execute a job or engage in any other intricate activity (Peter and Gomez, 2019).

Work skills can be categorized into two main types: technical (hard) skills and non-technical (soft) skills. Despite their apparent differences, both play crucial roles in achieving job success. Technical skills are tangible and job-specific talents that enable individuals to perform tasks more accurately and precisely. These skills pertain to the tools and machinery involved in the job, and individuals gain proficiency in them over time through practice. On the other hand, non-technical skills encompass interpersonal or people-related abilities that are applicable across various jobs. Examples of non-technical skills include communication, cooperation, and adaptability. They are often referred to as transferable skills because they can be effectively employed in multiple job settings (Dipesh Gurjar, 2018).

Numerous job advertisements (ads) are published daily, each targeting various job titles. The information contained within these job ads serves as the primary potential source for identifying the requisite skills candidates must possess. However, the unstructured nature of this data poses challenges in capturing crucial information effectively.

A persistent gap exists between the skills acquired by college graduates through educational systems and the actual skill sets demanded by real-world job opportunities. By analyzing recruitment data to extract the essential skills required for specific job positions, businesses and job seekers can better comprehend the alignment between human capabilities and career options.

The elimination of the skills gap holds significant importance in re-engaging individuals in the job market, reducing unemployment rates, and bolstering labor force participation. A key aspect of closing the skills gap involves accurately identifying the mismatch between the skills expected by employers and those possessed by job seekers. Once these skill gaps are identified, job seekers can receive recommendations for new skills or educational products to enhance their competitiveness in the job market and adaptability to evolving job requirements (Zhou et al., 2016).

Mismatches between education and labor market demands are a prominent subject of interest for job market research analysts (Wen and Maani, 2022; Serikbayeva and Abdulla, 2022). This complex issue can be examined from various angles, including the field of study, level of study, demography, job-specific competencies, design skills, and soft skills (Phaphuangwittayakul et al., 2018; Kabir, 2014).

Automatically extracting the necessary hard and soft skills from unstructured recruiting data, such as job descriptions and CVs, represents a critical initial step in addressing the aforementioned challenges (Cui et al., 2021).

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However, the development of skill extraction systems presents inherent difficulties due to existing challenges. The major issues encountered in the skill extraction task are as follows: (1) the scarcity of well-annotated data, and (2) the complexities in extracting newly emerging skills that were previously unknown.

Information technology, computer science, and related fields experience rapid skill appearance and disappearance over time. As such, staying abreast of these changes and preemptively understanding the required skills for a particular position can assist job seekers in preparing for a highly competitive job market.

This paper introduces a novel explainable ensemble learning model that combines transformers with Conditional Random Fields (CRFs) (Wallach, 2004) to extract both technical and non-technical skills from job descriptions. As no standard publicly available dataset exists for this task, a vast collection of job descriptions in English was scraped and annotated from prominent job boards such as LinkedIn, Glassdoor, Indeed, and StepStone. The dataset encompasses 22 job titles related to Information Technology (IT) across various English-speaking countries, and their corresponding job ads were considered for scraping and analysis.

The evaluation of the proposed model on a carefully curated dataset and its comparison with other recently introduced models designed for the same task reveal its superiority in both precision and recall metrics. Additionally, the results indicate that the model exhibits satisfactory performance in extracting soft skills when compared to using a dictionary lookup approach.

The proposed model combines the strengths of Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and the traditional Conditional Random Fields (CRF) algorithm. Furthermore, Bidirectional Encoder Representations from Transformers (BERT) are utilized to dynamically represent inputs for the deep models. The research objectives in this paper are threefold: (a) to assess the effectiveness of an ensemble model using contextualized word embeddings as input; (b) to investigate the impact of local features extracted by CNNs from raw inputs on the overall model performance; and (c) to determine if the proposed model can successfully extract novel skills not present in the training data.

Additionally, considering the significance of explainability in NLP tasks, the model's behavior is investigated using the LIME (Local Interpretable Model-agnostic Explanations) method (Mishra et al., 2017). This allows for a clear understanding of the reasoning behind the model's decision-making process.

Moreover, the contributions of this study can be listed as:

- **Hybrid Model Integration:** This research synergizes a conventional CRF model with a contemporary ensemble-based deep learning architecture to enhance skill extraction performance.
- **Feature Rich Input:** A diverse range of features is incorporated into the model to facilitate a comprehensive analysis of job descriptions.
- **Extensive Dataset Development:** A substantial and meticulously annotated dataset was curated to support robust model training and evaluation.
- **Explainable Ensemble Model:** Employing explainable NLP techniques to elucidate the decision-making processes within the ensemble model.

The proposed model achieved F-scores of 67% and 72% for extracting non-technical and technical skills, respectively. Compared to conventional CRF and hybrid deep learning models, the model demonstrated significant performance gains. Specifically, it surpassed these baselines by average margins of 10% and 6% for non-technical skills and 29% and 6.8% for technical skills, as measured by F-score.

The rest of the paper is organized as follows. We give a review of the related works in Section 2. In Section 3, the suggested model and specifics of its components are provided. Sections 4 and 5 describe the data preparation and the evaluation processes, respectively. Section 6 discusses the experiments and results achieved. In Section 7, model exaplainability is investigated. Finally, the paper draws conclusions in Section 8.

2. Related work

Skill extraction from textual resources is a subject of significant interest to both industry and academia, as evident from numerous research papers addressing the issue using various approaches. Previous works have primarily relied on text analysis and machine learning methodologies.

One of the pioneering models proposed for this purpose is presented in the study by Kivimäki et al. (2013). The authors introduced a two-phase method based on Wikipedia articles and their hyper-link graphs, supplemented with a set of skills from LinkedIn. In the first phase, they employ a vector space model or a topic model to analyze a query text and associate it with relevant Wikipedia articles. Leveraging these initial pages, they utilize the spreading activation algorithm on Wikipedia's hyperlink graph to identify articles corresponding to the LinkedIn skills.

In another approach, Smith et al. (2019), the authors propose a method for extracting relevant skill terms from job ads. They generate machine-labeled text from a small amount of human-labeled data, addressing the vocabulary problem and improving skill detection recall1. The approach analyzes syntactic patterns in job content to identify meaningful skills, emphasizing grammar-based cues rather than semantic information. This novel method aims to enhance skill extraction efficiency and accuracy.

Khaouja et al. (2021b), proposed an unsupervised method for skill extraction from real-world job advertisements. The study aims to automatically identify and extract hard skills (specific competencies) mentioned in job ads. Unlike manual annotation, this method operates in an unsupervised manner, making it scalable and adaptable to diverse job postings. The authors leverage word embedding models (e.g., Word2Vec) trained on a large collection of job ads. These embeddings capture semantic relationships between words, allowing for skill-related terms to be grouped together. The method clusters similar skills based on their co-occurrence patterns in job ads. By analyzing which skills frequently appear together, the approach identifies skill clusters.

Shi et al. (2020), introduced a market-aware skill extraction system. They employed two distinct data collection strategies: one involving experts in job postings and the other based on market signals. Their developed model, called *Job2Skill* incorporates deep learning techniques and market signals to model skill salience. Experimental results demonstrated that Job2skills significantly enhances the quality of multiple LinkedIn products, including job recommendation and skill suggestion systems. It showed +1.92% improvement in job apply rates for job recommendations (Jobs You May Be Interested In) and 37% reduction in suggestion rejection rate for skill suggestions provided to job posters.

Gugnani and Misra (2020), proposed a model for extracting explicitly expressed skills from job descriptions using natural language processing (NLP) techniques. Next, the approach identifies implicit skills—those not explicitly mentioned in a job description (JD) but contextually relevant based on geography, industry, or role. To infer implicit skills, the system finds other JDs similar to the target JD in a semantic space using a Doc2Vec model trained on 1.1 million JDs. Skills absent in the target JD but present in similar JDs are obtained, and their importance is weighted. Finally, various similarity measures are explored to match skills extracted from a candidate's resume to both explicit and implicit skills in JDs.

Bhola et al. (2020), introduced another deep learning-based skill extraction model, treating the problem as an extreme multi-label classification task with a wide range of skill class labels. Their skill extractor relies on language modeling approaches using transformers. Notably, this model is capable of recovering relevant skills that may be missing from job descriptions, employing a Correlation Aware Bootstrapping process to consider the structured semantic representation of skills and their co-occurrences.

Smith et al. (2021), presented an NLP-based model that extracts skills from text using syntactic headwords. They created an open vocabulary dataset of skills by utilizing a small set of human-created annotated data. The syntactic model can also be adapted for extracting entities of interest from other text domains, provided that the new text exhibits syntactic similarity to the texts used for training, as the method relies on syntactic dependency patterns to recognize skills.

One study that investigated the relationship between various artificial intelligence (AI) skills is conducted by Jia et al. (2018). To achieve this, the authors construct a knowledge graph representing these skills and employ a Long Short-term Memory Network (LSTM) to study their interactions. First, they create the knowledge graph using a rule-based approach based on AI skills prevalent in job markets. Next, the graph is visualized to reveal knowledge relationships. The study classifies AI jobs into two categories: general algorithm jobs and specific focus jobs. Notably, Python and Linux emerge as essential key skills across all AI roles. This analysis provides valuable guidance for job-seekers, companies, and universities, shedding light on the necessary skills in the dynamic field of artificial intelligence

Cao and Zhang (2021) worked on a model to comprehend the social requirements for job seekers interested in the position of "Data Analyst". The proposed model relies on deep learning techniques, combining LSTM and CRF models. Furthermore, Bidirectional Encoder Representations from Transformers (BERT) is employed to extract skill entities instead of the model's input layer.

Phaphuangwittayakul et al. (2018), investigated skill demand trends by analyzing job postings from various recruitment websites in Thailand. The study employs multiple techniques, including web scraping, keyword extraction, and visualization. By extracting relevant skills from job ads, the research sheds light on the skills most sought after by employers in the Thai labor market. Zhou et al. (2016), explored the application of Named Entity Recognition (NER) techniques for extracting skills from job postings. Specifically, it focuses on the German language context. NER plays a crucial role in identifying specific chunks of text as named entities (such as skills) and classifying them into predefined categories (e.g., "Person", "Organization", or "Location"). The method involves several steps, including tokenization, entity identification, classification, contextual analysis, and post-processing. By leveraging NER, researchers can transform unstructured job postings into structured information, facilitating data analysis, information retrieval, and knowledge graph construction.

Bhatia et al. (2019), proposed a project-oriented extractor for both technical and non-technical skills. For technical skills, they trained a naive Bayesian classifier on labeled data. To extract non-technical skills, the content of job descriptions is compared to a list of non-technical skills. Cui et al. (2021) suggested a deep learning-based model, combining Attention BiLSTM and CRF, to extract skills from internet recruitment data. They also investigated the impact of using Word2vec and BERT as model inputs. Nguyen et al. (2021) introduced a model to extract domain-specific skills trained with a small set of data from Japanese job descriptions. Their proposed model incorporates transfer learning with a stack of convolutional neural networks to learn hidden representations for classification.

Considering the significance of soft skills in the current job market, Khaouja et al. (2019), presented a methodology for generating soft skill taxonomies using a combination of DBpedia and word embedding. The combined model produces a set of related words that indicate how soft skills are listed in job advertisements. They also created a soft skills hierarchy by visualizing the taxonomy as a network and employing centrality measures.

Jiechieu and Tsopze (2021), demonstrated the effectiveness of using a convolutional neural network-based multi-label classification system to predict high-level skills from job seekers' resumes. They employed convolutional filters to extract low-level features, which serve as explanations for the target skills as classes.

In their work on automatic screening of job candidates' resumes, Pant et al. (2022) presented a parser model utilizing NLP techniques, word matching, and character positioning. Their focus was on extracting technical skills from a dataset comprising software engineering candidate requirements. The model generates a resume summary based on the extracted information and computes count scores for the detected skills, education, and experience levels.

Another skill extraction system for resumes was introduced by Sajid et al. (2022). Their suggested multi-stage architecture initially extracts text from various parts and blocks of a document, followed by skill extraction using named entity recognition and an enhanced ontology.

Ternikov (2022), investigated the demand for skills in the IT sphere within the Commonwealth of Independent States (CIS) region. The study aimed to discover the mapping between required skill sets and job occupations. To achieve this, the proposed methodology employs natural language processing, hierarchical clustering, and association mining techniques. By analyzing explicit information about the combinations of "soft" and "hard" skills required for different professional groups, the findings provide valuable insights for educational organizations, human resource specialists, and state labor authorities.

The Skill Preferences (SKiP) method, developed by Wang et al. (2022), constructs a model of human preferences and employs it to extract human-aligned skills from offline data. SKiP leverages human input to address downstream challenges using reinforcement learning after extracting human-preferred abilities.After extracting skills, SkiP ensures they align with human preferences. These human-preferred skills serve as behavior priors for downstream tasks. SkiP bridges the gap between data-driven skill extraction and human intent, substantially outperforming existing methods in robotic environments.

In their research, Takeuchi et al. (2023) presented a virtual reality system designed to identify expert-specific skills in a visual inspection task within a refinery setting. They introduced a Convolutional Neural Network (CNN) with the Class Activation Map technique, along with an explainable AI method, to analyze experimental data obtained from experienced and novice field operators. The primary objective was to determine the most significant contributors for classifying expert and novice behavior during 120-second inspections.

In addition to the works mentioned in this section, Khaouja et al. (2021a), Mashayekhi et al. (2024) and Xu et al. (2023) prepared a comprehensive survey that covers almost all relevant tasks related to skill extraction, examining numerous published studies in the field.

3. Proposed model

The primary objective of supervised learning algorithms is to explore a hypothesis space to find an appropriate hypothesis that can produce accurate predictions for a specific problem (Blockeel, 2011; Rokach, 2010; Yang et al., 2023). However, even if the hypothesis space contains well-suited hypotheses for a given problem, finding a good one can still be challenging. Ensembles aim to address this issue by combining multiple hypotheses to create a potentially stronger one. In this study, we propose an ensemble learning approach that utilizes multiple extractors to potentially achieve improved results. Ensembles of conventional and deep models offer several advantages (Yang et al., 2023). Combining these model types can leverage the interpretability and efficiency of traditional methods with the powerful feature extraction capabilities of deep learning. This synergy often results in enhanced predictive performance, reduced overfitting, and increased robustness to noise and outliers. Additionally, ensemble methods can mitigate the limitations of individual models, leading to more reliable and generalizable solutions. The different components of the proposed model are illustrated in Fig. 1. The specifics of all individual components of the model are elucidated below. Bidirectional Long Short-Term Memory (BiLSTM) (Ghaeini et al., 2018) and Conditional Random Fields (CRF) models (Wallach, 2004) are two distinct techniques employed for addressing sequence labeling problems, representing newer

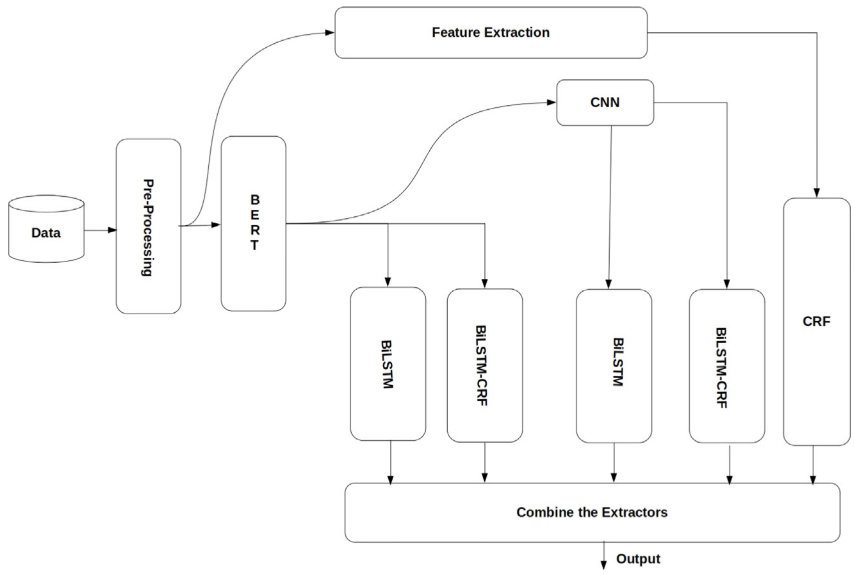

Fig. 1. Proposed model for skill extraction.

and older methodologies, respectively. In this study, both the BiLSTM and CRF models were utilized as primary solutions, independently or in conjunction with each other.

The proposed model comprises various modules. Initially, the input undergoes preprocessing to produce clean and tokenized text. BERT is then employed in two ways in the subsequent step to generate contextualized word embeddings for deep models. First, as a direct input for the BiLSTM and BiLSTM+CRF models. Second, as input to the CNN to extract local features. Convolutional Neural Network (CNN) is a Deep Learning algorithm capable of assigning importance (learnable weights and biases) to different aspects of the input through suitable filters. This allows the CNN to capture spatial and temporal dependencies in the input effectively. Due to the reduced number of parameters and weight reusability, the CNN architecture outperformed other deep architectures used on the provided input dataset, making it better at recognizing the input's complexity (Albawi et al., 2017). In the suggested model, the CNN's outputs have the same dimensions as its inputs.

Moreover, all deep models are trained in two ways: individually and in the context of multi-task learning (Zhang and Yang, 2018). While most machine learning applications focus on solving a single task at a time, multi-task learning aims to learn multiple tasks simultaneously while optimizing multiple loss functions. Instead of training separate models for each task, a single model is used to learn and complete all tasks simultaneously, using available data from various tasks to learn generalized data representations applicable in different contexts. In this study, multi-task learning is employed, where the tasks are identical, but the model architectures differ. Each task has its own loss function, denoted by L_i. The combined loss function, defined as the linear sum of the separate model loss functions used in our proposed model, is minimized (see Formula (1)).

$$\underset{\theta}{minimize} \sum_{i=1}^{M} L_i(\theta, Data) \tag{1}$$

where, M represents the number of extractors, which is set to 5. Additionally, θ denotes the parameters of the models. It is noteworthy that CRF differs from the other models as it is trained in isolation using distinct types of input. To arrive at the final decision for each potential skill, the outputs of various models are combined using majority voting.

For comprehensive information about the model's implementation, please refer to Section 6.

3.1. Model components - baselines

Our proposed model constitutes a composite of elementary extractors, and their outputs are combined using the majority voting scheme. One of the pioneering statistical models that have demonstrated effectiveness in addressing sequence tagging problems is **CRFs (Conditional Random Fields)** (Wallach, 2004). CRFs take context into account, meaning that when predicting something, the model considers the influence of neighboring samples by representing the prediction as a graphical model. Considering an input sequence denoted as $x = (x_1, \dots, x_m)$, where each element represents a word in a sentence, and a corresponding output sequence denoted as $s = (s_1, \dots, s_m)$, where each element represents a named entity tag, conditional random fields model the conditional probability $p(s_1, \dots, s_m \mid x_1, \dots, x_m)$.

To achieve this, we need to define a feature map function $\phi(x_1, \dots, x_m, s_1, \dots, s_m) \in R^d$, which maps the entire input sequence paired with the entire state sequence s to a d-dimensional feature vector. Subsequently, the probability can be modeled as a log-linear model with the parameter vector $w \in R$ according to Formula (2).

$$p(s \mid x; w) = \frac{\exp(w.\phi(x, s))}{\sum_{s'} \exp(w.\phi(x, s'))} \tag{2}$$

In this context, s' represents the set of all possible predictable sequences. The estimation of w is carried out utilizing the training data. Furthermore, the regularized log-likelihood function is precisely defined as Formula (3):

$$L(w) = \sum_{i=1}^{n} log p(s^i \mid x^i; w) - \frac{\lambda_2}{2}|w|_2^2 - \lambda_1 |w|_1 \tag{3}$$

where λ_1 and λ_2 are regularization hyperparameters to force the weight vector to be small as much as possible. At the conclusion of the process, the optimal values for w and s are computed using the following formulas, respectively: $\text{argmax}_{w \in R} L(W)$ and $\text{argmax}_s p(s \mid x; w^*)$.

Given the right features, it becomes feasible to achieve high-quality labeling. The Conditional Random Field (CRF) is notably flexible in terms of feature selection, and it does not necessitate conditional independence among features (Ponomareva et al., 2007). However, the primary drawback of CRF lies in the high computational complexity of the training step, making model retraining more challenging when additional training data samples become available. Furthermore, CRFs are unable to recognize unknown words, i.e., words that were not present in the training data sample.

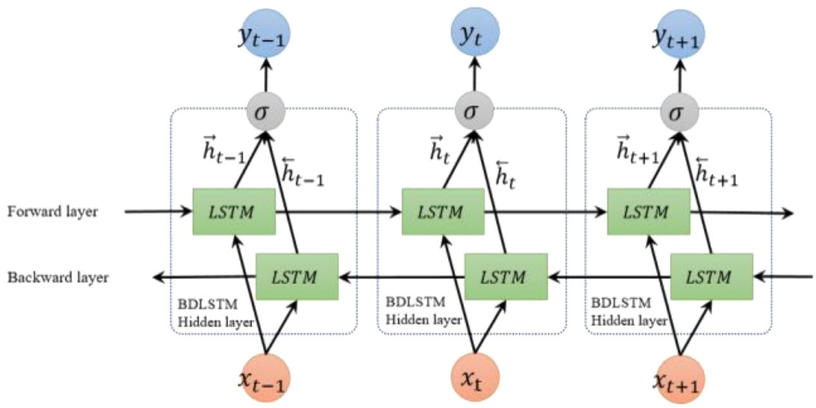

Fig. 2. Bidirectional LSTM.
Source: After Cui et al. (2018).

Recent advancements in deep learning have been addressing various sequence tagging problems. As a result, **BiLSTMs (Bi Directional Long Short Term Memory)** networks have been considered as an alternative baseline in this study. Fig. 2 illustrates the architecture, consisting of two LSTMs, each operating on the sequence in a distinct manner. Indeed, LSTM (Long Short-Term Memory) is a deep learning architecture utilizing an artificial recurrent neural network (RNN) with feedback connections, in contrast to conventional feed-forward neural networks (Hochreiter and Schmidhuber, 1997). When provided with a sequence of vectors $X = (x_1, x_2, \dots, x_n)$ as input, LSTM generates another set of hidden state vectors $H = (h_1, h_2, \dots, h_n)$ as output. The mathematical specifics of each LSTM cell are elucidated by Formula (4).

$$\begin{cases} i_t = \sigma(W_{xi}x_t + b_{xi} + W_{hi}h_{t-1} + b_{hi}) \\ f_t = \sigma(W_{xf}x_t + b_{if} + W_{hf}h_{t-1} + b_{hf}) \\ C_t = tanh(W_{iC}x_t + b_{iC} + W_{hC}h_{t-1} + b_{hC}) \\ o_t = \sigma(W_{io}x_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \\ c_t = f_t \mathbf{x} c_{t-1} + i_t \mathbf{x} C_t \\ h_t = o_t \mathbf{x} tanh(c_t) \end{cases} \tag{4}$$

In the BiLSTM model, at each time step t, x_t, h_t, and c_t represent the input, hidden state, and cell state, respectively. The input, forget, cell, and output gates are denoted as i_t, f_t, C_t, and o_t at time step t, where σ and tanh refer to the sigmoid and hyperbolic tangent functions, respectively.

The fundamental concept of BiLSTM networks involves presenting each training sequence both forwards and backwards to two separate recurrent neural networks. These networks are connected to the same output layer, granting the BiLSTM complete sequential information about all points before and after each point in the sequence. As the network can adaptively utilize the context information, there is no need to determine a task-dependent time-window or target delay size. The performance of the model is greatly influenced by having access to a large amount of annotated data and setting appropriate values for the deep models' hyperparameters.

In this study, the benefits of both CRFs and BiLSTM are combined in a linear scheme, resulting in a **BiLSTM-CRF** model. A linear layer is employed to compute scores associated with each label j at position t in the sequence, after extracting hidden vectors using BiLSTMs. In CRF, $w.\phi(x, s)$ scores how well the state sequence fits the given input sequence, denoted as $score_{crf}(x, s)$. In the BiLSTM-CRF, a non-linear neural network is introduced by combining BiLSTM and CRF as follows: $Score_{BiLSTM-crf}(x, s) = \sum_{i=0}^{n} W_{s_{i-1},s_i} \cdot BiLSTM(x)_i + b_{s_{i-1},s_i}$. Here, W_{s_{i-1},s_i} and b represent the weight vector and bias corresponding to the transition from state s_{i-1} to s_i, respectively. The scoring function allows for optimization of the conditional probability $p(s \mid x; W, b)$, following the same principles as in a traditional CRF, by propagating back through the network.

Additionally, to enhance the diversity of extractors, a Convolutional Neural Network (CNN) (Jacovi et al., 2018) is integrated into the suggested model. CNNs are multi-layered artificial neural networks capable of recognizing complex features in various input data types, including text and images. They consist of two fundamental layers: the *Convolution layer* for feature extraction from the data and the *Pooling layer* for condensing the feature map (Fig. 3). The convolution process extracts the most crucial features using kernels (filters), while pooling ensures feature detection throughout the network and further compresses the data sent to the CNN.

In all deep learning models, input data must be represented as numerical vectors. One common approach for encoding input text involves using either static word embeddings or context-sensitive embedding models. Static word embedding methods, such as Word2Vec (Church, 2017) and Glove (Pennington et al., 2014), assign the same vector to polysemous words without considering their context. On the other hand, dynamic (context-sensitive) models produce contextualized word representations. These models build upon the transformer architecture, which excels at capturing word associations in a sentence through attention mechanisms. Attention aids in alignment for sequence-to-sequence tasks, similar to how convolution aids in translation.

Numerous transformer-based language models have been developed for various languages, with the focus and amount of unstructured text used for training being the key distinguishing factors. In the proposed model, the inputs to the deep models are encoded using BERT (Kenton and Toutanova, 2019), which takes into account the context of each word. BERT is a state-of-the-art model designed to pre-train deep bidirectional representations from unlabeled text, conditioning on both left and right context in all layers. It utilizes the transformer, an attention mechanism that learns contextual relations between words or sub-words in a text. Unlike directional models that read text input sequentially (either left-to-right or right-to-left), the transformer encoder processes the entire sequence of words simultaneously, making it effectively bidirectional or, more precisely, non-directional.

4. Data preparation

An existing challenge in the skill extraction task is the absence of a standardized annotated dataset. Previous studies have resorted to using their own data for experiments due to the unavailability of such a dataset (refer to Section 2). To address this issue and align with our specific domain of interest, we collected and annotated data to create a dataset tailored for our research purposes.

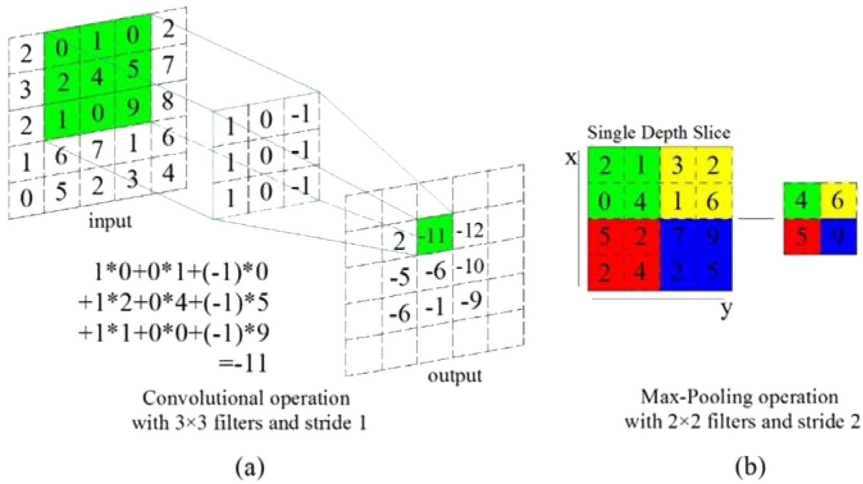

Fig. 3. Convolution and pooling operations.
Source: After Lin et al. (2019).

Table 1
Job roles are in demand.

Software engineer	UX designer	Front end developer
Database administrator	Machine learning engineer	Cloud architect
Business analyst	Mobile applications developer	DevOps engineer
Data scientist	Data analyst	Network engineer
Java developer	Data governance manager	BlockChain developer
Database developer	Data engineer	Big data architecture
System engineer	Back end developer	
Cyber security specialist	Data manager	

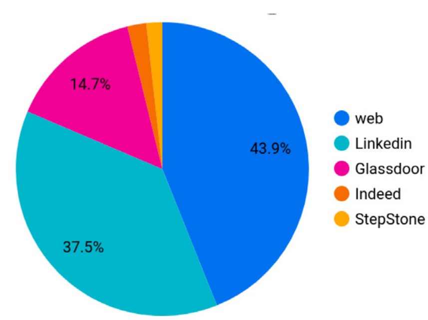

Fig. 4. Amount of data from an individual resource.

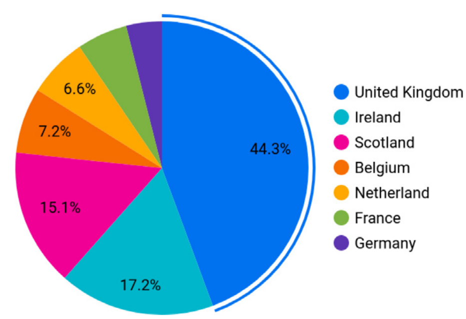

Fig. 5. Collected data per each country.

4.1. Data collection

In comparison to other fields, computer science and information technology-related jobs are in high demand by businesses and startups worldwide. We identified 22 job roles that exhibit significant demand, based on the number of job advertisements published on popular job boards in Europe during the final three months of 2021 (See Table 1).

In the subsequent step, job descriptions were scraped from four popular job boards (*Glassdoor, LinkedIn, Indeed, and StepStone*) between October and December 2022. To generate queries for job boards, the job roles listed in Table 1 were used along with various locations, including *United Kingdom, Netherlands, France, Ireland, Germany, Scotland, and Belgium*, and the language was set to English. These countries were selected as target locations due to their high demand for IT-relevant jobs, which ensures a larger pool of recruitment data, i.e., job descriptions.

In addition to job boards, web content was scraped using the Google Search API (Google team, 2021) to collect relevant free texts corresponding to the given query. Unlike job boards, which automatically return newly published job ads, using the Google Search API with the same query resulted in nearly identical outcomes.[1] To address this issue, the query was modified by incorporating additional key terms to the job roles, such as names of different cities and various years. For example, the query = *"machine learning engineer 2022 London"* was used as an example.

In total, 30,695 job descriptions and web documents were scraped from job boards and the web (using the Google Search API). The statistics on the collected data are illustrated in Figs. 4, 5, and 6. As observed from the figures, the amount of collected data is imbalanced concerning the platforms used for scraping and the target locations. However, as this study's primary focus is skill extraction from text rather than categorizing job advertisements based on their providers, the aforementioned data imbalance had minimal impact on the system's performance.

4.2. Data annotation

After data collection was completed, the duplicated and incomplete scraped ads were eliminated, resulting in 22,874 documents. Data screening at this stage involved applying specific criteria to remove irrelevant data, such as text lines with a length of less than 15 and those lacking certain keywords like "*Requirements*", "*Application*", "*Qualification*", "*Degree*", etc. During web surfing, a comprehensive list of both hard and soft skills was compiled, comprising 7170 and

[1] The following link presents an example page retrieved by the Google search api for *Machine Learning Engineer* position. https://www.prospects.ac.uk/job-profiles/machine-learning-engineer.

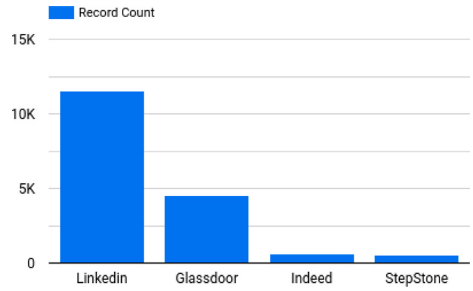

Fig. 6. Amount of data from different job boards.

Table 2
Properties of annotated dataset. B, I, H, and S stand for Beginning, Inner, Hard, and Soft respectively.

# sentences	207 327	# B-S-SKILL	103 083
# B-SKILL	510 949	# I-S-SKILL	290
# I-SKILL	172 826	# OUT	8 998 800
# B-H-SKILL	407 866		
# I-H-SKILL	172 536		

269 skills, respectively.[2] The large number of skills is attributed to considering the same skills in various written forms. Since these skills are expressed differently in writing, our skill set includes numerous duplicates (e.g., "Microsoft SQL Server", "SQL server", "Ms SQL", etc., all refer to the same ability). The annotators were tasked solely with determining the relevance of the 7170 skills to the selected jobs in this study. They collaborated and consulted the paper's author to resolve disagreements regarding skill relevance. After manually inspecting the hard skills, approximately 60% of them were filtered out, retaining only 2043 skills relevant to information technology.[3] Furthermore, the remaining skills exist in both single and multiple word forms.

The raw job descriptions underwent preliminary pre-processing, including the removal of redundant blank spaces, emojis, and the correction of contractions (Contractions are shortened words or word combinations, marked by apostrophes, formed by dropping letters. Eliminating contractions contributes to text standardization, which is valuable in textual data processing where words play a significant role). To address the challenges posed by nested skills, only those skills were considered for analysis, where none of their subsequences constituted another single-word skill. For example, in the case of *Microsoft SQL Server*, *SQL Server* or *SQL* would be disregarded as separate skills.

Next, all the documents were segmented into sentences using SPACY (Honnibal and Montani, 2017). Employing the IOB scheme (Krishnan and Ganapathy, 2005), the sentences were automatically labeled in two modes: firstly, both hard and soft skills were labeled as SKILL, and then they were labeled separately as H-SKILL and S-SKILL. Additionally, SPACY was used to tokenize the data for labeling purposes. The job descriptions and the skills were tokenized using the same tokenizer. For skills with more than two tokens, the header token was labeled B-X-SKILL, and the remaining tokens were labeled I-X-SKILL. When there was only one token, it was labeled as B-X-SKILL (where X represents H, S, or nothing corresponding to Hard-SKILL, Soft-SKILL, or just SKILL). After tokenization, discrepancies between the text and the skills were manually rectified. Sentences without any skills were removed from the dataset. Sentences lacking skills contribute to an increase in *OUT* tokens in the dataset, leading to a higher imbalance ratio between irrelevant entities and those of interest. The use of imbalanced data results in biased extractors with lower performance. The characteristics of the final dataset are presented in Table 2.

[2] The links below are of an example for skills resources: https://ec.europa.eu/social/main.jsp?catId=1326&langId=en
https://github.com/maciejszewczyk/linkedin-skills/blob/master/linkedin_skills.txt
https://www.developgoodhabits.com/soft-skills-list.

[3] The skills filtering was carried out under the supervision of two computer scientists holding PhD and Master degrees.

Table 3
Models' hyper parameters. In fact the kernel sizes used for CNN are 3 * 768, and 4 * 768. CRF_c1 = 0, Since the L2 regularization approach is only utilized for training CRFs, CRF_c1, CRF_c2 are considered 0 nd 1.0 respectively.

Hyper parameter	Value	Hyper parameter	Value
BiLSTM layers	2	Epochs	10
Units of LSTM	100	Dropout rate	0.25
CNN filters	16	CRF_c1	0.0
Optimizer	ADAM	CRF_c2	1.0
Learning rate	1e−5	Sequence length	100
CNN kernel sizes	3, 4	Pooling type	Maxpooling

5. Evaluation metrics

In simple terms, the skill extraction task involves tagging groups of words with labels like "SKILL" and "NON-SKILL". The evaluation of skill extraction systems commonly employs metrics such as recall, precision, and F1 score. Precision represents the ratio of correctly identified skills to all the skills recognized by the system, while recall is the ratio of correctly identified skills to the total number of actual skills present.

The performance of the system can be evaluated in two modes: exact matching and partial matching. Since this study aims to extract skills with complete representation, we utilize the exact matching approach to calculate the aforementioned scores based on Formula (5):

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP+FP}, \\ \text{Recall} &= \frac{TP}{TP+FN}, \\ \text{F1 Score} &= 2 * \frac{Precision * Recall}{Precision + Recall} \end{aligned} \tag{5}$$

where TP (True Positives) and FP (False Positives) represent the numbers of correctly and incorrectly tagged chunks of text as skills, respectively. FN (False Negatives) denotes the number of skills missed by the extractor.

6. Experiments and results

In this study, several experiments were conducted to gain a deeper understanding of the available methods. The configurations used to implement all components of the model, as presented in Section 3, are illustrated in Table 3.

In addition, all deep models utilized the *Categorical Cross Entropy* as the loss function, and the input consisted of 768-dimensional contextualized word embedding vectors produced by BERT.[4] The pretrained BERT model used in this study has 12 hidden layers (i.e., Transformer blocks), a hidden size of 768, and 12 attention heads (A).

This BERT model was pre-trained on English Wikipedia and BooksCorpus. The text inputs were normalized in a "cased" manner, preserving distinctions between lowercase and uppercase characters, as well as accent markers. Random input masking was independently applied to word pieces during training. The models were then trained using prepared training data in the subsequent step. Due to the lack of available test data to evaluate model performance, a 5-fold cross-validation approach was employed.

All experiments were conducted on a desktop computer (Intel Core I7–6900 K, 3.2 GHz, 64 GB RAM) equipped with an NVIDIA GeForce GTX 1080 TI GPU graphics card.

Table 4 presents a comparative analysis of various approaches for extracting hard and soft skills, evaluated using precision, recall,

[4] https://tfhub.dev/tensorflow/bert_en_cased_L-12_H-768_A-12/4.

Table 4
Results of the systems considering hard and soft skills separately. P, R, F are precision, recall, and Fscore, respectively. #S, show the number of skills which are correctly recognized.

	Soft skills				Hard skills			
	P	R	F	#S	P	R	F	#S
CRF	0.46	0.31	0.37	35 048	0.48	0.39	0.43	204 379
BiLSTM_1	0.64	0.51	0.56	54 633	0.68	0.56	0.61	296 350
BiLSTM_2	0.65	0.53	0.58	56 695	0.68	0.59	0.63	311 678
BiLSTM+CRF_1	0.69	0.57	0.62	60 818	0.70	0.63	0.66	321 897
BiLSTM+CRF_2	0.68	0.59	0.63	61 849	0.72	0.65	0.68	332 116
Ensemble model	**0.71**	0.63	0.66	64 942	0.75	**0.69**	0.71	347 445
Ensemble model_MT	**0.71**	**0.65**	**0.67**	65 973	**0.76**	0.69	**0.72**	352 554

Table 5
Results of different systems considering both hard and soft skills as SKILL. P, R, F are precision, recall, and Fscore, respectively. #S, show the number of Skills which are correctly recognized.

	SKILLs			
	P	R	F	#S
CRF	0.52	0.41	0.45	257 893
BiLSTM_1	0.66	0.54	0.59	343 857
BiLSTM_2	0.66	0.58	0.61	356 138
BiLSTM+CRF_1	0.71	0.61	0.65	380 699
BiLSTM+CRF_2	0.72	0.63	0.67	392 980
Ensemble model	0.78	0.67	0.72	423 682
Ensemble model_MT	**0.79**	**0.71**	**0.74**	435 962
Ensemble model_MT_excluding CRF	0.74	0.69	0.71	417 541

and F1-score. For comparative purposes, individual skill extractors derived from the proposed model served as baselines. A CRF model utilizing hand-crafted features was employed. BiLSTM models, designated as BiLSTM_1 and BiLSTM_2, were differentiated by their input sources: BERT embeddings for BiLSTM_1 and CNN features for BiLSTM_2. Subsequently, these BiLSTM models were integrated with CRF to create BiLSTM+CRF_1 and BiLSTM+CRF_2, respectively. All models were trained independently at this stage. The final two systems, Ensemble Model and Ensemble Model_MT, represent ensemble approaches, trained individually and collectively within a multi-task learning (MT) framework.

For all models, BERT weights remained fixed during fine-tuning, with a 30% validation split employed.

Furthermore, considering both hard and soft skills as SKILL, Table 5 presents a comparative performance analysis of the different models. These model configurations correspond to those outlined in Table 4. As anticipated, CRF exhibited the lowest performance due to its reliance on basic, hand-crafted features without advanced feature engineering. Notably, all models demonstrated superior performance in detecting hard skills compared to soft skills. This disparity can likely be attributed to the imbalanced distribution of hard and soft skill samples within the dataset.

The experimental results reveal a trade-off between precision and recall across all individual systems. Skill extraction as a Named Entity Recognition task presents unique challenges, with class imbalance being a primary contributor to low recall rates. Deep learning methodologies substantially improved upon the baseline CRF model in terms of both precision and recall. Additionally, incorporating CNN to capture local features prior to feeding BERT embeddings into BiLSTMs yielded marginal performance gains.

Ensemble models, regardless of training methodology, consistently surpassed the performance of individual models, as evidenced in Tables 4 and 5. The integration of multi-task learning models within the ensemble framework resulted in a slight performance enhancements over the standard ensemble approach, contributing to a more balanced precision–recall trade-off.

We conducted a 5 × 2cv paired t-test to verify the statistical significance of the F1 score improvement between the proposed model and the best-performing baseline (BiLSTM+CRF2). The 5 × 2cv paired t-test was proposed by Dietterich (1998) as a method for evaluating the performance of two models. In this test, the dataset is divided into two pieces repeatedly, resulting in five iterations. Each iteration involves splitting the data into a training set and a test set (50% training and 50% test data). Models are trained with the training split and evaluated against the test split in each iteration. The t statistic, which generally follows a t distribution with five degrees of freedom, is used to assess the null hypothesis that the proposed and baseline models perform equally well.

The computed p-values for Ensemble Model and Ensemble Model_MT, using the t statistic, are 0.0215 and 0.0342, respectively. By comparing these p-values with a pre-established significance level, $\alpha = 0.05$, we find that they are both smaller than α, allowing us to reject the null hypothesis and conclude that there is a significant difference between the two proposed models and the best-performing baseline.

Next, we manually compiled five new job descriptions for comparison with the results obtained by our model. An example job description of the selected ones is shown in Fig. 6. The yellow-highlighted spans of text represent all the skills identified by the human investigator.

The entities underlined in the text represent the ones that were not identified by the model, while the bold text spans indicate the false positives. The model demonstrates the ability to identify most of the skills determined by the human annotator. However, it struggles with identifying longer skills, indicating a weakness in capturing long dependencies within lengthy text.

Upon analyzing the false positives, it becomes evident that many of them were part of the skills used for data annotation, such as "data" or "flexibility". The proposed model achieves a precision of 0.78 and a recall of 0.73 for the identified skills when compared to human assessment, showcasing the close alignment of the results with the hypothesis. However, it also highlights the need for further improvement in the model's ability to accurately detect multi-word skills.

In the subsequent section, we undertake an examination of the model from the perspective of explainability.

7. Model's explainability

Many machine learning models, particularly those based on neural networks, can be considered as black boxes, providing an output without explicitly explaining the reasoning behind their decisions. Understanding and interpreting the decision-making process of these models is crucial. It is essential to closely scrutinize the model's output and, more importantly, be able to provide explanations for the decisions it makes (Søgaard, 2021; Gohel et al., 2021). By offering explanations for the model's output, we gain a higher level of confidence in either trusting or questioning the model's predictions.

In order to investigate the model's explainability, we made use of LIME to LIME (Mishra et al., 2017) is a model-agnostic technique, enabling its application to explain the output of any type of model without directly inspecting its internal structure. This is achieved by perturbing the local features surrounding a specific target prediction

....
Your role at Mapiq
As a key member of the **Data** team you will be responsible for developing new machine learning models for our **suggestion engine** and visualize new insights in our analytics platform. Where there is **data**, you will be involved to use that to smartly assist our customers and stay ahead of the competition. Your daily tasks will include:
Analyze, review and interpret complex data through the development of case studies, enabling the **data** to tell a story Build and maintain production-ready machine-learning models for our **suggestion engine** on our Azure cloud environment Be a storyteller to explain the 'why and how' of your **data**-driven recommendations to cross-functional teams and business owners Work on the business intelligence reports (PowerBI)
Who are we looking for?
We try to keep it realistic, but if we must describe our perfect candidate, it would be someone with 2+ years of experience as a Data Scientist, who understands Mapiq's mission to build great places to work.
Next to that you:
Are experienced in working with version management software (git) Experienced in working on cloud platforms (Microsoft Azure and Databricks are big plus) Hands-on experience of working with Python (e.g. TensorFlow, scikit-learn, NumPy, pandas, XGBOOST) Are analytical, accurate and structured, you can see projects through from start to completion Experience with Spark/PySpark is a **big plus** Experience with MLOps lifecycle principles; working experience with MLFlow is a **big plus**. Have a Bachelor's or Master's degree in **Computer Science**, Data Science, **Mathematics**, **Physics**, **Econometrics** or any other relevant beta studies Experience with building on business intelligence software: Power BI (or similar like Tableau) You are fluent in English, some Dutch would be great It would be even better if you can include some of the skills below:
Experience working with Data Lakehouse platforms such as DeltaTables, DeltaLiveTables Comfortable with DevOps practices (Git, CI/CD, Azure DevOps or similar) Knowledge of Restful APIs and microservices
What to expect?
The freedom to implement your own ideas in our product The flexibility of a hybrid working environment: work remotely as much as you want, and come to the office whenever you like to meet your team or visit the gym! We think work is about reaching goals and developing yourself professionally and personally. We care about your well-being and offer free mental health support. We have 100+ employees who have built a strong community of board game enthusiasts, sports fanatics, coffee lovers, and more! Be yourself and have fun!
... An example of a job description from which the suggested model has extracted the existing skills.

and observing the resulting model output. In our particular experiments, the tokens surrounding a target entity were modified, and subsequently, the model's output was assessed.

We considered a piece of text, *6+ years of Web UI/UX design experience Proven mobile web application design experience*, from a job description which has not been seen in the training data to conduct the explainability experiment.

In the context of LIME, perturbed instances are sampled around X, where X is the target entity of interest that we want to be explained, and their weights are assigned based on their proximity to X, with weight represented by size. The original model's predictions on these perturbed instances are obtained, and then a linear model is learned to approximate the behavior of the original model in the vicinity of X.[5]

As a result, LIME produces a list of tokens with contribution scores to the model's prediction, offering local interpretability. Additionally, this enables the identification of feature changes that will have the most significant impact on the prediction.

In an attempt to explain the label assigned to the word "UI/UX", the Lime outputs are presented in Fig. 7. The green color indicates that the token has a positive contribution to the predicted label, while the red color, represented by "<BIAS>", indicates a negative contribution. The model made a correct prediction with a probability of 0.95, classifying "UI/UX" as part of a multi-token skill labeled as I-SKILLS. The word "Web" was a significant indicator supporting this prediction. However, the label B-SKILLS had a lower probability of 0.018, and the word "Web" had a strong negative contribution, aligning with the correct classification of the model.

While LIME is a valuable tool for interpretability, it is not without limitations. Its reliance on locally approximating complex models can lead to instability in explanations, as minor data perturbations may result in significantly different interpretations. Additionally, LIME's focus on individual instances might obscure global patterns within the model, hindering a comprehensive understanding of its decision-making process. Furthermore, the quality of LIME explanations is sensitive to

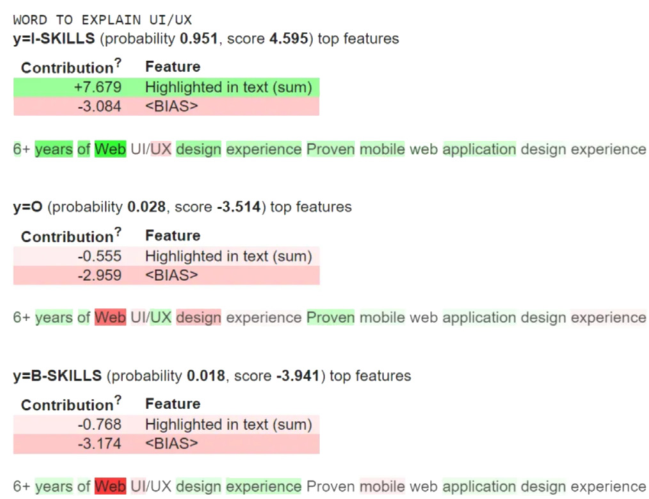

Fig. 7. Lime's output.

hyperparameter choices, requiring careful tuning for optimal results. These factors necessitate cautious interpretation of LIME-generated explanations and highlight the need for complementary interpretability techniques.

8. Conclusion

This paper introduces a novel skill extractor system based on an ensemble learning approach, which combines the traditional CRF algorithm with deep learning architectures. The contextualized word embedding produced by BERT, a state-of-the-art transformer for language modeling, serves as input for the deep models. Additionally, a CNN layer is inserted between BERT and the deep models' input to extract local features and enhance the diversity within the pool of extractors. A CRF model with simple hand-crafted features is also included in the pool.

[5] We made use of ELI5 library to automate the data sampling and explanation process, https://eli5.readthedocs.io/en/latest/.

To address the lack of a standard publicly available annotated dataset for this task, a specialized dataset is created by scraping job descriptions from job boards, focusing on 22 IT and computer science-related job roles. The dataset is manually annotated with sets of hard and soft skills.

The experimental results, along with the significance test, demonstrate that the ensemble learning approach outperforms other skill extraction systems in terms of performance. Moreover, training deep models in a multitask learning scheme improves the system's performance while reducing the gap between recall and precision.

Moreover, through the utilization of the local interpretable model-agnostic approach, we have demonstrated that the model's decisions are rational, primarily based on the importance of the surrounding words around the skill terms and phrases. This indicates that the model comprehends the context in which both hard and soft skills appear.

Future steps to enhance the system's efficiency include increasing the size of the extractor pool and employing well-engineered features for the CRF models.

CRediT authorship contribution statement

Abbas Akkasi: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The author declare that there is no conflict of interests regarding the publication of this paper.

Data availability

All data collected and used for the experiments will be available upon reasonable request.

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